



GIBBS DISTRIBUTION

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► **Theorem:** For $\beta \geq 0$ define π_β as the probability measure with density $\pi_\beta(x)$ over $\mathrm{d}x$

$$\pi_\beta(x) = \frac{\exp(-\beta U(x))}{Z(\beta)} \eta(x), \quad Z(\beta) = \int \exp(-\beta u) \eta(x) \mathrm{d}u$$

- Where there is a unique $\beta \geq 0$ such that $\pi_\beta[U] = u$

$$\text{minimize :} \quad \text{KL}(\pi||\eta) = \pi \left[\log \frac{d\pi}{d\eta} \right]$$

$$\text{Constraint :} \quad \pi[1] = 1, \quad \pi[U] = u$$

▶ We refer to π_β as the Boltzmann distribution and $\beta \mapsto Z(\beta)$ as the partition function where

- Since π_β is an exponential family, we have $\log Z(\beta)$ is convex, and satisfies

$$\frac{d}{d\beta} \log Z(\beta) = -\pi_\beta[U] := -\langle U \rangle_\beta, \quad \frac{d^2}{d\beta^2} \log Z(\beta) = \pi_\beta[(U - \pi_\beta[U])^2] := I(\beta)$$

▶ Note that when $\eta(dx) = dx$ we have $\beta \geq 0$ but we can have $\beta < 0$ if $\langle U \rangle_\beta < \eta[U]$

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- Note that when $\eta(\mathrm{d}x) = \mathrm{d}x$ we have $\beta \geq 0$ but we can have $\beta < 0$ if $\langle U \rangle_\beta < \eta[U]$

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